

Module	Week	Session #	Session Title	Learning Objectives	Date	Start Time	End Time	Faculty Type	To be taken by
Module 1: Analytics Foundations + GenAI + Spreadsheets	1	1.1	Statistics: Understanding Data and Averages	Numerical vs categorical data and how analysts use each; Mean; median; and mode to summarize business metrics like sales and ratings; Why outliers distort the mean and when median is a more honest business number; Range as a quick measure of how spread out performance is				Professor	
Module 1: Analytics Foundations + GenAI + Spreadsheets	1	1.2	Analytics Workflow, Metrics & KPIs	Understand what an analytics workflow is; List the steps in an analytics workflow (problem → data → analysis → insight); Break down a business problem into smaller questions; Map a simple problem to each step of the analytics workflow; Understand what metrics and KPIs are; Identify metrics for simple business scenarios; Differentiate between a metric and a KPI; Convert a business question into a measurable KPI					
Module 1: Analytics Foundations + GenAI + Spreadsheets	1	1.3	GenAI for Analytics: Prompt, Check, Improve	Understand what GenAI is in the context of analytics; Identify where GenAI can be used in an analytics workflow; Write basic prompts for simple analytics tasks; Use simple prompting patterns (clear instruction, expected output format); Generate structured outputs (tables, lists) using prompts; Interpret outputs generated by GenAI; Identify incorrect or low-quality GenAI outputs; Apply simple validation checks to verify outputs					
Module 1: Analytics Foundations + GenAI + Spreadsheets	2	2.1	Clean Up the Data	Load datasets into spreadsheets (Excel/Sheets); Identify common data issues (missing values, duplicates, formatting issues); Clean datasets by removing duplicates and fixing formats; Sort and filter data to inspect it					
Module 1: Analytics Foundations + GenAI + Spreadsheets	2	2.2	Make Data Ready for Analysis	Structure datasets into consistent formats (columns, data types); Fix inconsistent entries (text, numbers, dates); Validate cleaned data using simple checks; Prepare a clean dataset ready for analysis					
Module 1: Analytics Foundations + GenAI + Spreadsheets	3	3.1	Formulas for Analysis	Use basic spreadsheet formulas (SUM, AVERAGE, COUNT); Apply formulas across rows and columns; Create new columns using formulas; Perform simple descriptive analysis using formulas					
Module 1: Analytics Foundations + GenAI + Spreadsheets	3	3.2	Pivot Tables and Quick Insights	Create pivot tables from datasets; Use pivot tables to summarize data; Compare categories using pivot tables; Extract simple insights from pivot outputs					
Module 2: SQL for Data Analysis	4	4.1	Statistics: Spread; Variability and Outliers	Why two datasets with the same average can behave very differently; Range; variance; and standard deviation to assess consistency and risk; Identifying outliers and how they affect business conclusions; Variability as a signal of reliability in performance data				Professor	
Module 2: SQL for Data Analysis	4	4.2	SQL Query Basics	Understand how data is stored in tables (rows and columns); Write basic SELECT queries; Select specific columns from a table; Filter data using WHERE conditions					
Module 2: SQL for Data Analysis	4	4.3	Sorting and Filtering in SQL	Sort data using ORDER BY; Combine filtering and sorting in one query; Retrieve specific subsets of data; Write simple readable SQL queries					
Module 2: SQL for Data Analysis	5	5.1	Aggregation Essentials	Use aggregation functions (SUM, COUNT, AVG); Calculate totals and averages from data; Understand how aggregation works on datasets					
Module 2: SQL for Data Analysis	5	5.2	Grouping for KPIs	Group data using GROUP BY; Compute KPIs using grouped data; Compare grouped results across categories; Interpret aggregated outputs					
Module 2: SQL for Data Analysis	6	6.1	Joining Tables Together	Understand why joins are needed; Identify common columns between tables; Combine tables using JOIN; View combined data from multiple tables					
Module 2: SQL for Data Analysis	6	6.2	Insights from Combined Data	Analyze data after joining tables; Compare values across tables; Identify patterns in combined datasets; Extract insights from joined data					
Module 2: SQL for Data Analysis	7	7.1	Subqueries in Action	Write subqueries inside SQL queries; Use subqueries to filter data; Break complex queries into smaller steps					
Module 2: SQL for Data Analysis	7	7.2	CTEs and GenAI for SQL	Write queries using CTEs; Structure complex queries for better readability; Use GenAI to generate SQL queries; Validate query outputs for correctness					
Module 3: Tableau Dashboards + Storytelling	8	8.1	Statistics: Probability and Uncertainty	Probability scale from 0 to 1 and expressing it as a percentage; Likelihood of business events such as a customer purchasing or churning; Simple probability calculations like conversion rate and click-through rate; Why probability is not certainty and how to communicate that to stakeholders				Professor	
Module 3: Tableau Dashboards + Storytelling	8	8.2	Tableau Basics and First Charts	Understand the Tableau interface; Load data into Tableau; Create basic charts (bar, line); Map fields to axes and visuals					
Module 3: Tableau Dashboards + Storytelling	8	8.3	Choosing the Right Chart	Create different types of charts; Choose the right chart for the data; Compare multiple chart types for the same dataset; Improve chart readability using labels and formatting					
Module 3: Tableau Dashboards + Storytelling	9	9.1	Building a Dashboard	Create a dashboard in Tableau; Add multiple charts to a dashboard; Arrange charts within a dashboard layout					
Module 3: Tableau Dashboards + Storytelling	9	9.2	KPIs and Trends on One View	Add KPIs to dashboards; Show trends using visualizations; Compare segments within dashboards; Use dashboards to support analysis					
Module 3: Tableau Dashboards + Storytelling	10	10.1	Dashboard Design Basics	Apply design principles to dashboards (clarity, layout); Reduce clutter in dashboards; Organize visuals for better readability					
Module 3: Tableau Dashboards + Storytelling	10	10.2	Dashboards That Support Decisions	Improve dashboards for decision-making; Highlight key insights visually; Align visuals to tell a clear story; Review dashboards for clarity and usefulness					
Module 3: Tableau Dashboards + Storytelling	11	11.1	Reading Dashboards for Insight	Read and interpret dashboard outputs; Identify trends and patterns in visual data; Convert charts into insights					

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Module 3: Tableau Dashboards + Storytelling	11	11.2	Insight Writing with GenAI	Use GenAI to draft insight summaries; Generate narrative explanations from dashboards; Review and correct generated narratives; Refine insights for clear communication					
Module 4: GenAI for Analytics Workflows + Basic Python	12	12.1	Statistics: Relationships in Data	Correlation to identify patterns between business variables like ad spend and sales; Positive and negative correlations and what they signal; Weak vs strong relationships and how to read them; Why correlation does not imply causation and how to avoid misleading business conclusions				Professor	
Module 4: GenAI for Analytics Workflows + Basic Python	12	12.2	Summarise Compare and Report with GenAI	Write prompts for summarising and comparing data using GenAI; Extract key points and identify important information from GenAI outputs; Generate structured reports (tables, summaries) using GenAI prompts; Improve outputs by refining prompts					
Module 4: GenAI for Analytics Workflows + Basic Python	12	12.3	Breaking Down the Workflow	Break an analysis problem into multiple steps; Define steps required in an analytics workflow; Design simple GenAI-supported workflows					
Module 4: GenAI for Analytics Workflows + Basic Python	13	13.1	Multi-Step GenAI Workflows	Combine multiple steps into a workflow; Use GenAI across different stages (analysis, reporting); Add validation steps in workflows					
Module 4: GenAI for Analytics Workflows + Basic Python	13	13.2	Python Fundamentals I	Work with variables and basic data types (integers, floats, strings, booleans); Use conditional statements (if/elif/else) to make decisions based on data					
Module 4: GenAI for Analytics Workflows + Basic Python	14	14.1	Python Fundamentals II	Work with lists, tuples, and dictionaries; Write for loops and while loops to iterate over lists, ranges, and dictionaries					
Module 4: GenAI for Analytics Workflows + Basic Python	14	14.2	Python Fundamentals III	Define and call functions with parameters and return values; Read and write files using Python (open, read, write); Understand common file formats (CSV, text) relevant to data work					
Module 4: GenAI for Analytics Workflows + Basic Python	15	15.1	Python for APIs and GenAI Integration	Make HTTP requests using the requests library; Understand API structure (endpoints, headers, JSON responses); Call the OpenAI API using Python to send prompts and parse responses; Build a simple script that combines file reading, functions, and API calls					
Module 4: GenAI for Analytics Workflows + Basic Python	15	15.2	Getting Started with Python Data	Understand how Python is used for data analysis; Load datasets using pandas; View and inspect datasets using head and info; Select and filter rows and columns in a DataFrame					
Module 4: GenAI for Analytics Workflows + Basic Python	16	16.1	Cleaning Data with Pandas	Clean data using pandas; Modify columns and values; Handle missing values and duplicates in pandas; Perform simple transformations on datasets					
Module 4: GenAI for Analytics Workflows + Basic Python	16	16.2	Visualizing Data with Python	Create basic plots using matplotlib (line, bar, scatter); Customize charts with titles, labels, and colors; Plot directly from pandas DataFrames; Compare Python-based visualization with Tableau for different use cases					
Module 4: GenAI for Analytics Workflows + Basic Python	17	17.1	Connecting the Tools	Combine outputs from spreadsheets, SQL, Tableau, and Python; Connect different tools into one workflow; Organize analysis steps end-to-end					
Module 4: GenAI for Analytics Workflows + Basic Python	17	17.2	End-to-End Analytics Workflow	Demonstrate a complete analytics workflow; Use GenAI to document analysis steps; Generate and communicate final insights clearly					
Module 4: GenAI for Analytics Workflows + Basic Python	17	17.3	Statistics: Distributions; Insights and Drawing Conclusions	What data distributions are and how the shape of data changes what it means; How skewed data makes averages misleading in business reports; How to read histograms to spot trends; gaps; and unusual values; How to draw fair conclusions from data instead of guessing				Professor	
Capstone Project	18	18.1	Capstone Task	Capstone Task					
Capstone Project	18	18.2	Capstone Task	Capstone Task					